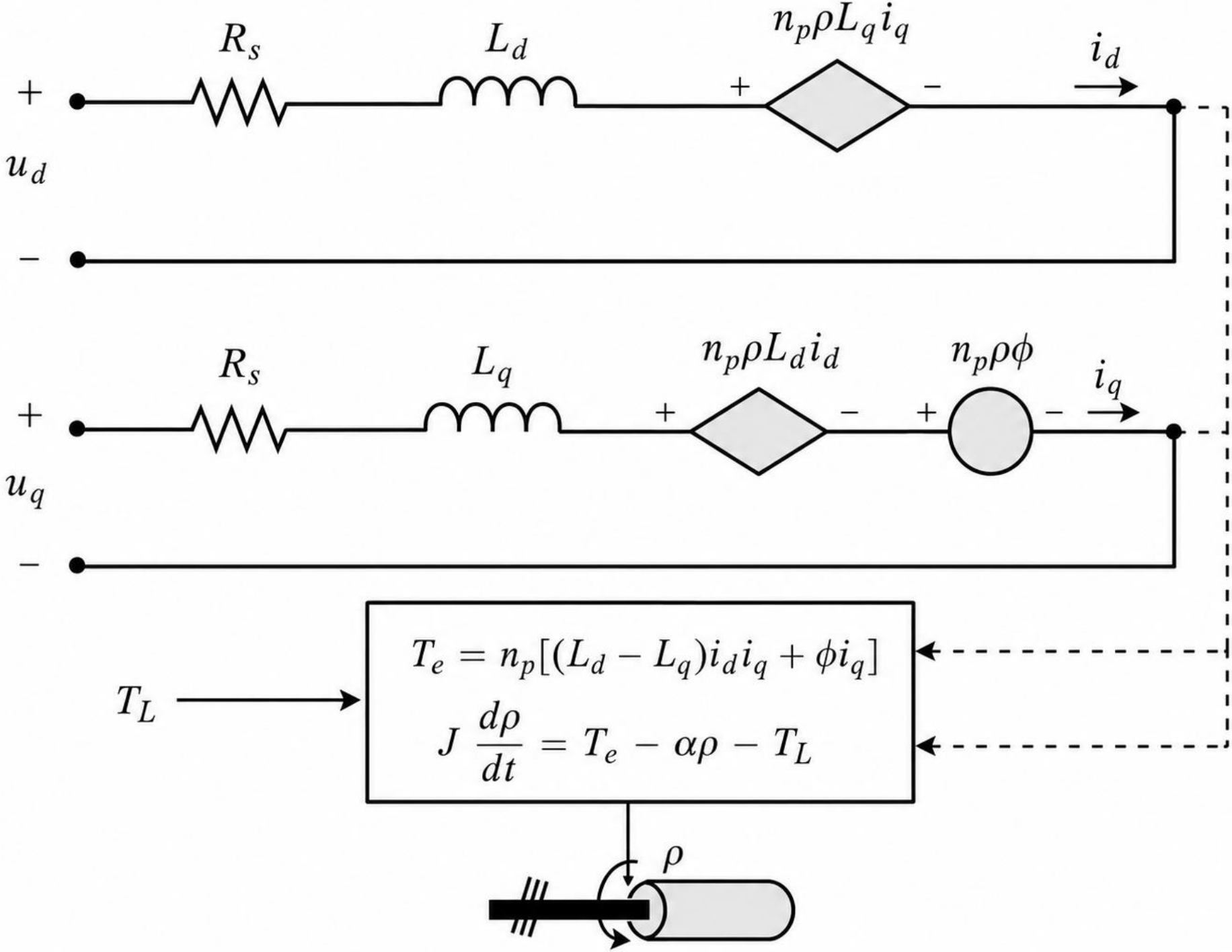

R_s
L_d
$n_p\rho L_q i_q$
i_d
+
u_d
−
R_s
L_q
$n_p\rho L_d i_d$
$n_p\rho\phi$
i_q
+
u_q
−
T_L
$T_e = n_p[(L_d - L_q)i_d i_q + \phi i_q]$
$J\frac{d\rho}{dt} = T_e - \alpha\rho - T_L$
ρ